AgriDirect AI

AI-Powered Crop Demand Prediction and Direct Farmer-to-Company Marketplace

Project Requirement Document (PRD)

Document version	1.0 (Draft for developer review)
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Prepared by	Onkar (Project Lead / Client Liaison)
Prepared for	Development team
Project type	Web platform + AI/ML module (MVP first, then scale)
Status	Requirements baseline - open items listed in Section 15

1. Project Overview

AgriDirect AI is a web and mobile-friendly platform that does two things for farmers. First, it uses AI to predict local crop demand and price direction so a farmer can decide what to grow and when. Second, it lets farmers list their expected or ready produce so that companies and bulk buyers can discover them and place requirements directly, reducing dependence on the traditional trader/mandi chain.

The product combines two concepts: (a) an **AI Crop Demand Predictor** (decision support using mandi, weather and soil data) and (b) a **Direct Farmer-to-Company Marketplace** (listing, matching and deal flow).

Core value statement

The platform tells the farmer which crop is likely to be in demand locally, and after harvest connects the farmer directly with interested companies and buyers.

Overall flow

Stage	What happens
1. Farmer input	Farmer registers and enters crop details: location, land area, crop, expected quantity, expected harvest date.
2. Data signals	System pulls weather, soil and mandi (price/arrival) data for that location and season.
3. AI demand prediction	Model estimates local demand level, demand trend and price direction for the crop.
4. Crop recommendation	Ranked shortlist of suitable crops for the farmer's place and season, with a simple explanation.
5. Requirement matching	Farmer's produce listing is matched with company requirements by crop, quantity, location and date.
6. Direct deal	Company contacts farmer / sends an offer; both negotiate and confirm the deal on the platform.

2. Problem Statement

- Farmers decide what to sow with little visibility into local demand, mandi prices, weather and soil conditions. Information is scattered across mandis, traders and informal networks.
- When many farmers follow the same signal they grow the same crop; oversupply pushes prices down at harvest time.
- After harvest, farmers depend on the traditional chain (local trader, commission agent, mandi) and have no easy way to find bulk buyers who actually need their produce.
- Companies (food processors, exporters, retailers, feed manufacturers) struggle to find reliable supply near them in the quantity and timeframe they need.
- Farmers need simple, timely guidance in their own language, not another complex dashboard.

3. Goals and Success Criteria

Business / product goals

- Give farmers a clear, easy-to-read crop and demand recommendation before sowing.
- Create a direct discovery channel between farmers and bulk buyers.
- Deliver an MVP that can be demonstrated end-to-end in one pilot region.

Measurable targets for MVP (to be confirmed with client)

Metric	Target
Demand prediction quality	Documented validation metrics (MAPE / RMSE for price, accuracy / F1 for demand class) reported by region and season; baseline model beaten
End-to-end flow	Farmer listing to company requirement match to deal confirmation works without manual intervention
Languages	English + Hindi + Marathi in MVP
Page / API performance	Key pages load in under 3 seconds on a 4G connection; prediction API under 2 seconds
Pilot	At least 1 region (e.g., one district in Maharashtra) and 3-5 crops live

4. Stakeholders and User Roles

Role	Description	Key needs
Farmer	Smallholder or mid-size grower; may have low digital literacy and use a basic Android phone.	Simple UI, local language, crop advice, find buyers, safe deals.
Company / Buyer	Processor, trader, exporter, retailer, FPO, or institution needing bulk produce.	Post requirements, discover nearby supply, compare listings, contact farmers.
Admin	Platform operator (client team).	Verify users, manage crops/data, monitor deals, view analytics, handle disputes.
Data / ML engineer	Maintains data pipelines and models.	Retrain models, monitor accuracy, manage data sources.

5. Scope

In scope - MVP (Phase 1)

- User registration and login (farmer, company, admin) with OTP-based phone verification.
- Farmer profile, land and crop details.
- Data ingestion for mandi prices/arrivals, weather and soil data.
- AI crop demand and price-direction prediction plus ranked crop recommendation.
- Farmer produce listings and company requirement postings.
- Matching engine (rule-based scoring) between listings and requirements.
- In-platform enquiry / offer / accept-reject deal flow and notifications.
- Multilingual interface (English, Hindi, Marathi).
- Admin dashboard with verification, monitoring and basic analytics.

Out of scope for MVP (Phase 2 / future)

- Online payments, escrow and invoicing.
- Logistics / transport booking and tracking.
- IoT sensor, drone and satellite data integration.
- Government scheme and procurement integration; export market signals.
- Native iOS / Android apps (MVP is a responsive web app / PWA).
- Contract farming and credit / insurance features.

6. Functional Requirements

Priority: **M** = Must have (MVP), **S** = Should have, **C** = Could have (later).

ID	Requirement	Pri.
FR-01	Users can register with mobile number and OTP; select role (farmer / company). Email optional for companies.	M
FR-02	Farmer profile: name, village/district/state, GPS or map-pin location, land area, irrigation source, soil type (if known), preferred language.	M
FR-03	Company profile: company name, type, GST/registration number (for verification), operating locations, contact person.	M
FR-04	Admin can verify or reject company (and optionally farmer) accounts; only verified companies can post requirements.	M
FR-05	Farmer can request crop advice by entering location, land area and season; system returns ranked crops with demand level, trend and price direction.	M
FR-06	Each recommendation shows a plain-language explanation (e.g., 'Demand expected to rise because local arrivals are falling and rainfall is favourable').	M
FR-07	Farmer can create a produce listing: crop, variety, quantity (tonnes/quintals), expected ready date, expected price (optional), location, photos (optional).	M
FR-08	For each listing the system displays AI insight: local demand, demand trend and nearby company requirement (aggregate).	M
FR-09	Company can post a requirement: crop, quantity, required-by date, delivery location, quality specification, price offer or range.	M
FR-10	Matching engine ranks listings against requirements by crop, quantity, distance, date and quality; both sides see suggested matches.	M
FR-11	Company can search and filter available produce (crop, quantity, district, ready date, distance) and contact the farmer or send an offer.	M
FR-12	Deal workflow with statuses: Enquiry, Offer sent, Negotiation, Accepted, Rejected, Completed, Cancelled. Both parties can view history.	M
FR-13	Notifications by SMS / WhatsApp and in-app for new matches, offers and status changes.	S
FR-14	Multilingual UI (English, Hindi, Marathi) and multilingual advisory text; language selectable at any time.	M
FR-15	Market dashboard: price trend charts and arrival trend for selected crop and mandi.	S
FR-16	Voice input / read-aloud advisory for low-literacy users.	C
FR-17	Ratings and reviews after deal completion.	S

ID	Requirement	Pri.
FR-18	Admin dashboard: user management, listings and requirements moderation, deal monitoring, data pipeline status, model performance.	M
FR-19	Report/flag listing or user; admin review workflow.	S
FR-20	Export of deal history and listings to CSV/PDF for users and admin.	C

7. Worked Example (Reference Scenario)

A farmer grows **10 tonnes of maize**, ready in 20 days. The platform should behave as follows:

Step	Platform behaviour
Farmer posts listing	Maize, 10 tonnes, ready in 20 days, district X.
AI insight shown	Local demand: High. Expected demand: Increasing. Nearby company requirement: 8 tonnes.
Company side	Verified company sees the farmer's available produce in search/matches and sends: 'We require 8 tonnes of maize.'
Farmer response	Farmer accepts, counter-offers on price, or rejects. Remaining 2 tonnes stay listed for other buyers.
Outcome	Farmer discovers a potential bulk buyer directly instead of relying only on the traditional chain.

Positioning line for demos: *"We don't only predict what farmers should grow - we connect them with who needs it."*

8. AI / ML Requirements

8.1 Models

Model	Purpose	Suggested approach
Price / demand forecaster	Predict local demand level and price direction (up / stable / down) for 2-12 weeks ahead per crop and market.	Gradient boosting (XGBoost / LightGBM) as baseline; time-series models (Prophet / SARIMAX / LSTM) for comparison. Choose by validation.
Crop recommender	Rank suitable crops for a location and season.	Hybrid: agronomic suitability (soil, rainfall, season) combined with predicted demand/price score.
Matching score	Rank listing-to-requirement pairs.	Rule-based weighted score in MVP (crop match, quantity fit, distance, date fit, quality); learn weights from deal outcomes later.
Explanation layer	Generate simple local-language advisory.	Feature-importance (SHAP) mapped to templated sentences; optional LLM for phrasing, translated to Hindi/Marathi.

8.2 Data sources

Data	Source (to be validated by developer)	Notes
Mandi prices and arrivals	Agmarknet (agmarknet.gov.in), data.gov.in open datasets	Daily; check API / scraping permissibility and rate limits.
Weather and forecast	India Meteorological Department; open weather APIs (e.g., Open-Meteo, NASA POWER)	Historical + forecast by lat/long.
Soil	Soil Health Card data, ISRIC SoilGrids, ICAR datasets	District / grid level.
Crop calendar and yield	Government crop statistics, ICAR, state agriculture department	Sowing/harvest windows by region.
Company requirements	Collected live on the platform	Used for local demand signal and matching.

8.3 ML engineering expectations

- Data pipeline (scheduled daily) for ingestion, cleaning, feature engineering and storage; failures must be logged and alerted.
- Validate models by region and season using time-based splits (no random leakage). Report metrics in a model card.
- Provide a confidence indicator with every prediction; never present a forecast as a guarantee.
- Handle cold start (new crop/region with little data) with fallback to state-level or crop-level averages.
- Model versioning, scheduled retraining (at least monthly) and drift monitoring.
- Prediction served via a REST endpoint; response time under 2 seconds (cache common queries).

9. System Architecture and Suggested Tech Stack

The stack below is a recommendation. The developer may propose alternatives with justification, provided they meet the requirements in this document.

Layer	Suggestion
Frontend	React or Next.js, Tailwind CSS; responsive and installable as PWA; i18n library for English / Hindi / Marathi.
Backend API	Python FastAPI (preferred, shares language with ML) or Node.js (NestJS/Express). REST with OpenAPI docs.
Database	PostgreSQL (with PostGIS for distance queries); Redis for caching and queues.
ML / data	Python, pandas, scikit-learn, XGBoost/LightGBM, Prophet/statsmodels; MLflow or simple model registry; Airflow or cron/Celery for scheduled jobs.
Auth	OTP login (e.g., Firebase Auth or MSG91/Twilio) + JWT sessions; role-based access control.
Notifications	SMS / WhatsApp Business API provider, email (SES/SendGrid), in-app.
Maps / location	Google Maps or OpenStreetMap/Leaflet for location pin and distance.
Hosting	AWS / GCP / Azure or Indian cloud; Docker containers; CI/CD via GitHub Actions.
Monitoring	Sentry for errors, basic logging/metrics, uptime alerts.

Logical components

Component	Responsibility
Web / PWA client	Farmer, company and admin interfaces.
API gateway / backend	Auth, business logic, listings, requirements, deals, notifications.
Data ingestion service	Fetch and clean mandi, weather and soil data on schedule.
ML service	Training, evaluation, prediction and explanation endpoints.
Matching service	Scoring and ranking listings vs requirements.
Database and cache	Operational data, features, predictions, audit logs.

10. Core Data Entities (Indicative)

Entity	Key fields
User	id, role, name, phone, email, language, status (pending/verified/blocked), created_at
FarmerProfile	user_id, village, district, state, lat, lng, land_area, irrigation, soil_type
CompanyProfile	user_id, company_name, type, registration_no, locations, verified_by, verified_at
Crop	id, name (en/hi/mr), category, season, duration_days
Listing	id, farmer_id, crop_id, variety, quantity, unit, ready_date, expected_price, lat, lng, status
Requirement	id, company_id, crop_id, quantity, unit, required_by, delivery_location, price_min, price_max, quality_spec, status
Match	id, listing_id, requirement_id, score, status
Deal	id, listing_id, requirement_id, quantity, agreed_price, status, history[]
MarketPrice	id, crop_id, mandi, date, min/max/modal price, arrivals
WeatherRecord / SoilRecord	location key, date, parameters
Prediction	id, crop_id, region, date, demand_level, trend, price_direction, confidence, model_version
Notification, Review, AuditLog	Standard fields; AuditLog records every admin and deal-status action.

11. Indicative API Endpoints

Endpoint	Purpose
POST /auth/otp/request, /auth/otp/verify	Login / registration via OTP
GET/PUT /farmers/me, /companies/me	Profile management
POST /advice/crops	Ranked crop recommendations for location and season
GET /predictions/demand?crop=&region;=	Demand level, trend, price direction, confidence
GET /market/prices?crop=&mandi;=	Historical price and arrival series
POST/GET/PUT /listings	Farmer produce listings
POST/GET/PUT /requirements	Company requirements
GET /matches?listing_id= | requirement_id=	Suggested matches
POST /deals, PATCH /deals/{id}/status	Create deal / offer and update status
GET /admin/users, PATCH /admin/users/{id}/verify	Admin verification and moderation

12. Non-Functional Requirements

Area	Requirement
Usability	Mobile-first, large touch targets, minimal text, icons plus local language; works on low-end Android devices.
Performance	Pages under 3 s on 4G; prediction API under 2 s; support 1,000 concurrent users in MVP.
Availability	99% uptime target in MVP; daily database backups with tested restore.
Security	HTTPS everywhere; hashed secrets; role-based access; input validation; protection against OWASP Top 10; rate limiting on OTP and APIs.
Privacy	Collect minimum personal data; store phone numbers and documents securely; consent for data use; comply with India's DPDP Act, 2023 requirements.
Scalability	Stateless backend, containerised, horizontally scalable; background jobs via queue.
Reliability	Data pipeline retries, alerting on failures, graceful fallback if external data APIs are down.
Localisation	All UI strings externalised; Unicode (Devanagari) fonts; date/number formats for India; units in quintal/tonne.
Maintainability	Clean modular code, README, API docs (OpenAPI), unit tests (target 70%+ on core logic), linting, environment-based config.
Compliance and trust	Disclaimer that predictions are advisory, not guaranteed; Terms of Use and Privacy Policy pages; platform acts as a connector, not a party to the trade (to be confirmed with client's legal advisor).

13. Suggested Milestones and Deliverables

Timeline is indicative and should be refined by the developer after review. Assumes one full-stack developer plus one ML developer (or one developer covering both).

Phase	Duration	Deliverables
0. Discovery and design	1 week	Requirement sign-off, UI wireframes (key screens), finalised stack, data-source feasibility check.

Phase	Duration	Deliverables
1. Foundation	2 weeks	Project setup, DB schema, auth/OTP, roles, farmer and company profiles, admin verification.
2. Data and AI	3 weeks	Data pipelines, cleaned datasets, baseline and improved models, validation report, prediction and recommendation APIs.
3. Marketplace	2-3 weeks	Listings, requirements, matching engine, deal workflow, notifications.
4. Frontend and multilingual	2 weeks (parallel with 3)	Responsive UI, advisory screens, dashboards, English/Hindi/Marathi.
5. Testing and pilot	2 weeks	QA, security checks, UAT with sample farmers and companies, bug fixing, performance tuning.
6. Deployment and handover	1 week	Production deployment, documentation, admin training, source code and credentials handover.

Indicative total: 10-12 weeks for MVP. Weekly progress demo and written status update expected every Friday.

14. Acceptance Criteria

- All 'Must have' functional requirements (FR marked M) are implemented and demonstrated.
- A complete flow works on staging: farmer registers, gets crop advice, posts listing, verified company posts requirement, match is suggested, offer is made and deal is accepted.
- Model validation report delivered with metrics by region and season and comparison against a naive baseline.
- No critical or high-severity open bugs; medium bugs documented with fix plan.
- App tested on at least 3 Android devices (including a low-end one), Chrome and Safari.
- Source code in client-owned repository, with README, setup guide, API documentation and environment configuration.
- Deployed to production environment with backups and monitoring configured.

15. Assumptions, Risks and Open Items

Assumptions

- MVP is a responsive web app / PWA; native apps follow later.
- Public / open data sources are sufficient for MVP-level prediction accuracy in the pilot region.
- Initial pilot covers one region and 3-5 crops.
- Payments, logistics and legal contracts happen outside the platform in MVP.

Key risks

Risk	Mitigation
Mandi / government data is incomplete, delayed or hard to access via API.	Early feasibility spike in Phase 0; cache and fall back to state-level averages; consider manual data upload by admin.
Low prediction accuracy for some crops or regions.	Report confidence; limit MVP to crops with sufficient data; always show advice as guidance, not certainty.
Cold start: not enough companies or farmers on the platform.	Seed with pilot partners (FPOs, local processors); admin can create requirements on behalf of companies.
Low digital literacy of farmers.	Simple UI, local language, voice support, assisted onboarding by field agents.
Fake listings or requirements, deal disputes.	Verification of companies, reporting and moderation, audit logs, ratings.
Scope creep.	Change-request process; anything outside Section 5 'In scope' is estimated separately.

Open items to confirm with client before development starts

#	Question	Owner	Status
1	Final budget, timeline and payment milestones.	Onkar / Client	Open
2	Pilot region and the first 3-5 crops.	Client	Open
3	Languages required beyond English, Hindi and Marathi.	Client	Open
4	Business model: free, commission per deal, subscription for companies?	Client	Open
5	Legal position of platform (connector vs trading party); terms and privacy policy ownership.	Client	Open
6	Branding: logo, colours, domain name; use of the name 'AgriDirect AI'.	Client	Open
7	Hosting preference and who owns cloud accounts and API keys (SMS, maps, WhatsApp).	Client / Developer	Open
8	Availability of any existing company or FPO data for the pilot.	Client	Open
9	Data-source licensing / terms-of-use check (Agmarknet etc.).	Developer	Open

16. Working Agreement with Developer

- Single point of contact for requirements and approvals: Onkar.
- Weekly demo and status report; blockers raised within 24 hours.
- Code in a shared private Git repository from day one; regular commits and pull requests.
- Any change in scope, timeline or cost is documented and approved in writing before work begins.
- All source code, data, models, documentation and credentials are handed over at project completion.
- Developer to review this document and reply with questions, estimate and any proposed changes within 3 working days.

End of document. Version 1.0 - to be updated after developer review and client confirmation of open items.